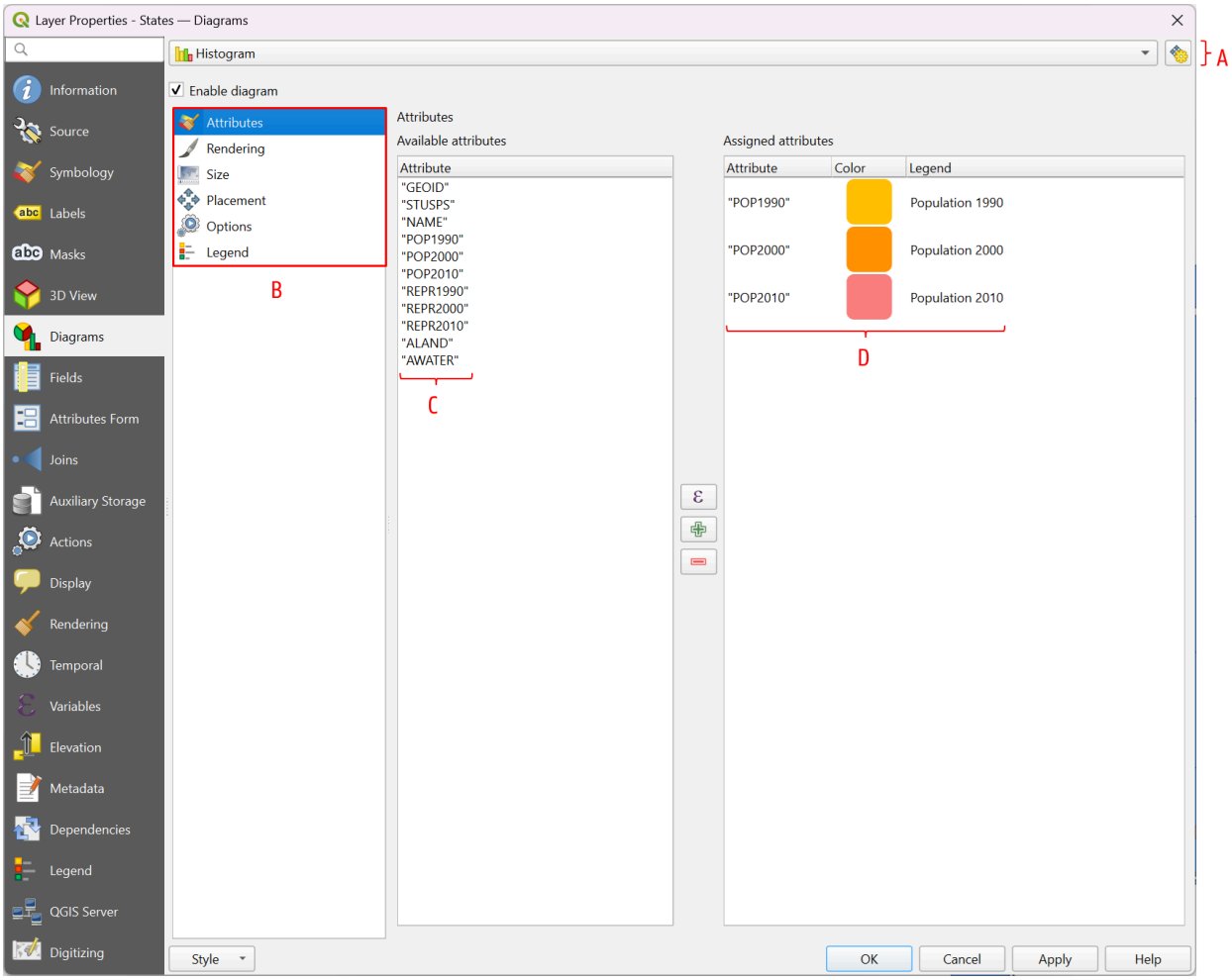
4.2.6 Diagrams

In QGIS you can show information on your map not only through the colour, size or shape of the features themselves, but also with diagrams.

Configuring diagrams

You set up diagrams in the *Diagrams* tab of the *Layer Properties* window. Do not forget to tick *Enable diagram* at the top of this tab, otherwise the diagrams are not shown on the map canvas.

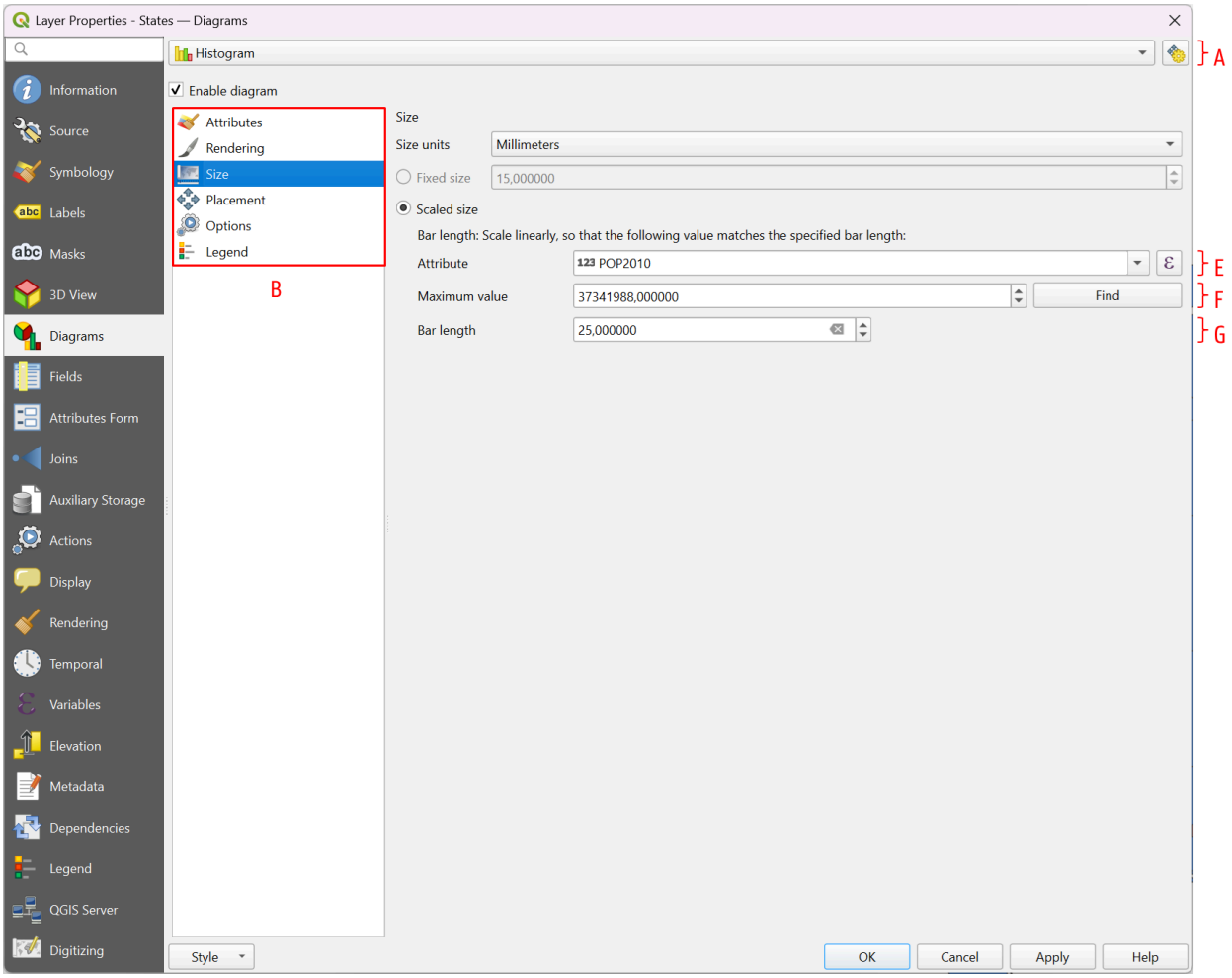


At A you choose the **diagram type**. In QGIS 4 these are *Pie Chart*, *Text Diagram*, *Histogram*, *Stacked Bars* and *Stacked Diagram* (an interface to add several diagrams at once). The diagram settings are spread over several **tabs** (B).

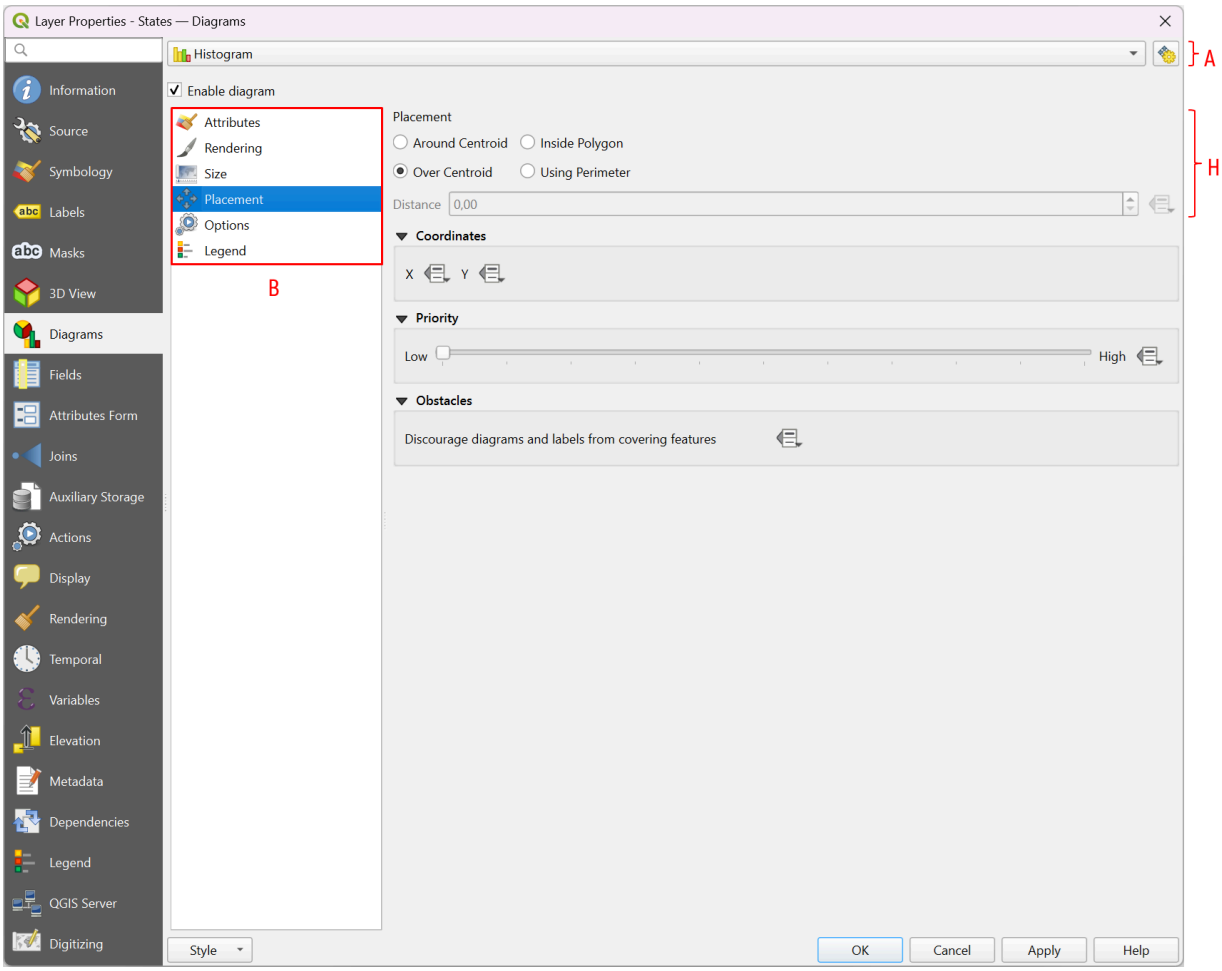
- In the **'Attributes'** tab you choose which attribute fields the diagram shows. Only **numeric fields** can be used. To add fields, select them at C and click the green plus symbol. Each added field receives a random colour. To change it, double-click the colour at D.
- In the **'Rendering'** tab you set basic display options, such as the bar width of a histogram, the border style of the diagram…
- In the **'Size'** tab (see the image below) you set how large the diagrams are drawn. For histograms, for example, this is the length of the bars, and for a pie chart the diameter. The size is either fixed (*Fixed*

size) or scaled (*Scaled size*).
With a **scaled size**, the size of each diagram depends on the value of an attribute field that you choose (E). Under *Size* or *Bar length* (G) you set the maximum size of the diagram (by default in millimetres). Under *Maximum value* (F) you can type the value that corresponds to this maximum size, or let QGIS look up the largest value of the field for you (click *Find*).

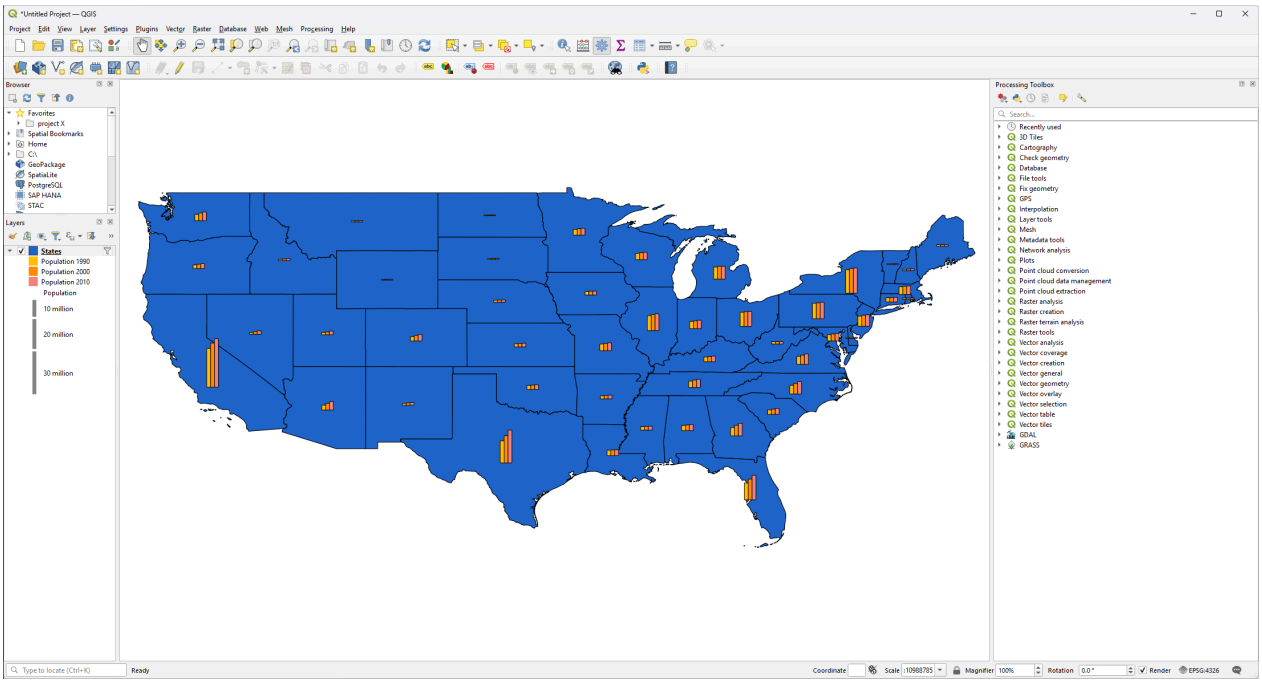
> Warning
>
> If the *Maximum value* at F is smaller than the real maximum in the data, some diagrams are drawn **larger than the maximum size** at G.



- In the **'Placement'** tab you configure how each diagram is positioned on the map in relation to its feature (H). With the *Over* options (*Over point*, *Over line*, *Over centroid*), the diagram is centred on the point, the middle of the line or the centroid of the polygon. With the *Around* options, QGIS draws it next to that spot, at the distance you set under *Distance*, for example so that it does not cover a point symbol. For lines, you can also choose whether the diagram is placed above, on or below the line. Polygon layers have two more options: *Inside Polygon* and *Using Perimeter* (on the border of the polygon).



- In the **'Options'** tab you find a few settings that only apply to certain diagram types. For a *Histogram*, you can for example configure the direction of the bars under *Bar Orientation* (*Up*, *Down*, *Right* or *Left*).
- In the **'Legend'** tab you choose whether the diagram gets a legend and how it should look. The legend explains the colours, the size, or both. The labels themselves can be updated via the *Attributes* tab, at D (see above). The image below shows the result: a histogram per state on the map (population in 1990, 2000 and 2010), and a legend in the *Layers* panel that explains the colours and the bar length.


*Untitled Project — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Favorites
project X
Spatial Bookmarks
Home
C:\
GeoPackage
SpatiaLite
PostgreSQL
SAP HANA
STAC
Layers
States
Population 1990
Population 2000
Population 2010
Population
10 million
20 million
30 million
Processing Toolbox
Search...
Recently used
3D Tiles
Cartography
Check geometry
Database
File tools
Fix geometry
GPS
Interpolation
Layer tools
Mesh
Metadata tools
Network analysis
Plots
Point cloud conversion
Point cloud data management
Point cloud extraction
Raster analysis
Raster creation
Raster terrain analysis
Raster tools
Vector analysis
Vector coverage
Vector creation
Vector general
Vector geometry
Vector overlay
Vector selection
Vector table
Vector tiles
GDAL
GRASS
Type to locate (Ctrl+K)
Ready
Coordinate
Scale 10988785
Magnifier 100%
Rotation 0.0 °
Render
EPSG:4326